THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 1: **Exploring the Investigative World of Science** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A student tests the effect of the mass of a bob on the time period of a pendulum. Which is the dependent variable? 1

(a) mass of the bob **(b)** length of the string
(c) time period **(d)** place of testing

2. After heating, the temperature readings are 40, 60, 80 and 100 degrees C at times 1, 2, 3 and 4 min. The rate of rise is: 1

(a) 40 degrees C per minute **(b)** 100 degrees C per minute
(c) 10 degrees C per minute **(d)** 20 degrees C per minute

3. A data table has more than one outlier. The best action is to: 1

(a) check the experimental method **(b)** stop recording
(c) use the biggest **(d)** drop all of them

4. In the above test, which must be kept constant? 1

(a) length of the string **(b)** number of masses
(c) mass of the bob **(d)** time period

5. For a study with the dependent variable 'height of a plant', the most suitable unit is: 1

(a) cm **(b)** degree C **(c)** s **(d)** kg

6. The time for an object to fall decreases when it is dropped from a lower height. The variables that are related are: 1

(a) mass of earth and shape **(b)** height and time
(c) colour and time **(d)** shape and colour

7. A student gets times of 12.1 s, 11.9 s, 12.0 s for 20 swings. The mean is: 1

(a) 36.0 s **(b)** 12.1 s **(c)** 12.0 s **(d)** 11.9 s

8. Which of the following is a controlled variable when comparing the cooking of puris at 150 and 180 degrees C? 1

(a) colour at the end **(b)** puffing time
(c) oil temperature **(d)** thickness of dough

9. To test whether the type of soil affects plant growth, which factor should be varied? 1

(a) amount of water **(b)** seed type **(c)** type of soil **(d)** amount of light

10. If 20 swings take 36 s, the time period of one swing is: 1

(a) 18 s **(b)** 0.18 s **(c)** 1.8 s **(d)** 720 s

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why are the independent and dependent variables different in an investigation? 2

12. State two limitations of using only the senses for observation. 2

13. What is an outlier? How should it be handled? 2

14. Calculate the mean of the readings 4.2 s, 4.0 s, 4.1 s and 4.3 s. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A student plans to test whether sugar dissolves faster in warm water. List the variable to change, the variable to measure and three things to keep same. 3

16. A student measured the length of a table with a hand span and found 8; another got 9. Explain the difference and suggest a better method. 3

17. Discuss how curiosity helped in discovering a common phenomenon such as the boiling of milk, giving the questions raised. 3

OR

The readings of a pendulum for 10 swings are 18.2 s, 18.0 s and 18.4 s. Find the mean time and the time period. 3

18. Why must the experimenter note the units of each quantity in a table? 3

OR

Explain why many experiments need a time limit and fixed procedure. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Plan an investigation to find out which of three liquids (water, milk, oil) cools fastest. Mention what you measure, what you keep constant, and the procedure. 5

OR

A report states 'The more fertiliser we gave the taller the plants got.' Write the variables, the likely hypothesis and ways to test it fairly. 5

20. Marks of a pendulum experiment: length (cm) 20, 40, 60 and time period (s) 0.9, 1.3, 1.6. Describe the relationship and a conclusion. State two variables to keep constant. 5

OR

Why do scientists record negative or unexpected results too? Give three reasons. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND